Yoni Nachmany
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Machine learning and software engineer proficient in Python and deep learning and motivated by frontier AI models. I thrive in small, ambitious, and collaborative teams translating research innovations into real-world applications.

Experience

Zeus AI — Remote
Head of Data Engineering, First Hire — May 2024 – May 2025

Zeus AI spun out of NASA to design novel AI weather models with Earth observations. I developed and managed software, infrastructure, and workflows for generating pretraining data, evaluating new datasets, and serving models.

- Productionized PyTorch foundation model for global weather forecasting 50x faster than traditional models, applying Apple's Massively Multimodal Masked Modeling transformer architecture to remote sensing data.
- Built robust real-time data processing and inference pipelines to accurately predict weather variables with low latency, integrating AWS SNS/Lambda triggers and provider API calls with Modal web endpoints and H100s.
- Crawled and tokenized TB-scale spatiotemporal data with Slurm and MPI on high performance computing.
- Deployed models and efficiently served hourly predictions of changes in solar energy for traders with FastAPI.

The Linux Foundation — Remote (Contract)
Senior Software Engineer, Data Pipeline Task Force — March 2024 – May 2024

Overture Maps Foundation is an open data collaboration under Linux with member companies like Microsoft, Meta, and Amazon. I contributed to the GA launch of the Global Entity Reference System for geospatial data conflation.

- Prototyped Parquet release changelogs with Spark/Athena and evaluated partitioning to optimize throughput.

Gro Intelligence — New York City
Software Engineer, Data Platform Project Lead — April 2023 – March 2024

Gro Intelligence built an agricultural data platform with predictive models mapping the impact of climate change on commodities. I led the development of large-scale ETL workflows that enabled training of machine learning models.

- Orchestrated Airflow DAGs to compute a century of climate projections with ECS in a cloud data lake on S3.
- Contributed to and oversaw the development of internal Python library to standardize usage of Xarray/Zarr.

The New York Times — New York City (Contract)
R&D Engineer — December 2021 – September 2022

The NYTimes R&D Team explores emerging technologies and develops new technical capabilities and story formats. I contributed to 3D mapping research and owned the CV/NLP work to create a more accurate digital news archive.

- Scaled transcription of scanned newspapers with multimodal APIs and evaluation framework for 10M articles.
- Implemented 3D satellite stereo reconstruction with photogrammetry libraries to support newsroom coverage.

Mapbox — Washington, D.C.
Software Engineer — September 2019 – November 2021

Mapbox's Imagery team maintains and improves the basemap by sourcing, updating, and enhancing provider data. I extended the capabilities of the data processing pipeline to handle new sources and derive value-added products.

- Launched 3D map product with 135M km^2 of satellite imagery and saved $1M by processing open aerial data.
- Improved the Mapbox Satellite UX for users loading data across zoom level groups with novel fusion algorithm.
- Implemented and evaluated deep learning-based object detection/super-resolution (EDSR) for map customers.

Education

University of Pennsylvania — Philadelphia
Master of Science in Engineering (MSE), Data Science — May 2019
Bachelor of Science in Engineering (BSE), Networked and Social Systems Engineering — May 2018

Hunter College High School — New York City

Teaching

TA for NETS 213: Crowdsourcing & Human Computation with Prof. Chris Callison-Burch	Spring 2019
TA for ESE 305: Foundations of Data Science with Prof. Victor Preciado	Fall 2017

Publications

Global Atmospheric Data Assimilation with Multi-Modal Masked Autoencoders	ICDM 2025
Learning Word Ratings for Empathy and Distress from Document-Level User Responses	LREC 2020
Design Considerations for High Impact, Automated Echocardiogram Analysis	ICML 2020
Mitigating Geographic Bias of Image Classifiers with Multilingual Image Data	D4GX 2019
Detecting Roads from Satellite Imagery in the Developing World	CVPR 2019
Generating a Training Dataset for Land Cover Classification to Advance Global Development	NeurIPS 2018
Decoded: Crowdsourcing Summaries of Internet Privacy Policies	HCOMP 2017

Fellowships/Internships

Data Science for Social Good Fellow at Imperial College and The Alan Turing Institute	Summer 2019
Machine Learning Intern at Radiant Earth Foundation	Summer 2018
Inaugural Earth on AWS Fellow at Sinergise (acquired by Planet)	Spring 2018
Open Source Software Fellow at Azavea	Summer 2017
Software Engineer Intern at Meetup	Summer 2016
Machine Learning Research Intern and hackNY Fellow at Clarifai	Summer 2015

Talks

Generating Elevation Data from Stereo Pairs of Satellite Imagery	GeoPython 2023
Depth Perceptions - Perspectives on 3D & Digital Twins	SatSummit 2022
Mapping Historical "Street View" Images of New York City: Visualizing Geotagged Archival Photos	FOSS4G 2022
Under-Mapped Spaces: New Methods and Tools for Critical Storytelling	Stanford 2022
Deep Transfer Learning for Land Cover Classification on Open Multispectral Satellite Imagery	FOSS4G 2019